Pivotal Tokens Are Mostly Uncertain Tokens

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Abstract

Pivotal Token Search (PTS) labels a generated token as pivotal when it moves the model's probability of reaching a correct answer by more than a threshold, estimated by sampling many rollouts and adjudicating each with an oracle. We ask what that expensive label is actually measuring. Using a reimplementation of the search that batches rollouts across queries—and that emits exact token positions rather than positions recovered by re-tokenizing—we generate pivotal-token events for Qwen3-0.6B over 3,000 GSM8K questions, and train linear probes to predict, from the residual stream at position $t-1$, whether the token at t will be pivotal. The probe reaches **0.855 AUROC** (95% CI $[0.799, 0.903]$, clustered by question), and beats a token-identity lookup table by a wide margin (paired difference $+0.191$, CI $[+0.116, +0.258]$). But it does *not* separate from next-token uncertainty: entropy alone reaches 0.814, and the paired difference is $+0.039$ with a confidence interval spanning zero. We argue this is close to definitional rather than incidental—a token can only move the success probability if the continuation was not already determined—and that it connects the Phi-4 pivotal-token line of work directly to the "forking token" account of critical positions. We also report a negative result on the sharper question: whether the *sign* of the impending shift is decodable, which at this scale it is not (0.614, CI $[0.482, 0.735]$). Finally we document two ways this measurement is easy to get wrong, both of which we initially got wrong ourselves: collapsing a question's sampled rollouts to one canonical sequence discards 74% of the supervision, and selecting the best of 29 layers on the reporting split inflates the headline.

1 Introduction

A language model solving a multi-step problem does not distribute its risk evenly across the tokens it emits. The Phi-4 technical report [Abdin et al., 2024] formalized this as *Pivotal Token Search* (PTS): a token t is pivotal when appending it to a prefix moves the estimated probability of eventually solving the task past a threshold,

$$|\Delta p| = \big|P(\text{success} \mid \text{prefix} + t) - P(\text{success} \mid \text{prefix})\big| > \tau. \tag{1}$$

Estimating either term requires sampling many completions and adjudicating each with an oracle, so PTS is an offline instrument that costs tens of rollouts per candidate position.

Our starting question was whether that instrument can be amortized: is pivotality linearly decodable from the residual stream at $t-1$, before the token is produced? A probe read one step early is a different object from one read at the pivotal token itself—the latter is a post-hoc classifier, the former is available while there is still a decision to influence.

It is. But the more informative result is *what else* predicts it. The question is only interesting if the answer is not already known for a cheaper reason, and two cheaper reasons exist. *Token identity*: some tokens may simply be pivotal more often, in which case a lookup table suffices and no representation is implicated. *Next-token uncertainty*: high-entropy "forking" tokens are the incumbent notion of a critical position [Wang et al., 2025].

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We find the probe clears the first comfortably and does not clear the second. On our data, next-token entropy predicts PTS's own label at 0.814 AUROC—within noise of a trained probe on the full residual stream. We take this seriously rather than treating it as a nuisance: it suggests that what PTS measures is, to a first approximation, where the model is uncertain, and that the two literatures have been describing overlapping phenomena with different instruments.

Our contributions:

- Evidence that pivotality is linearly decodable at $t-1$ (0.855 AUROC) and is *not* explained by token identity (Section 5.1).
- Evidence that it is *not separable* from next-token uncertainty, using a paired bootstrap over questions rather than the overlapping-interval comparison that would have hidden it (Section 5.2).
- A negative result on the signed question—whether the direction of the impending shift is decodable—at this scale (Section 5.3).
- A rebuilt labelling pipeline, and a characterization of two failure modes in measuring any of this: branch collapse, which discards 74% of the available supervision, and layer selection on the reporting split (Sections 3.1 and 5.4).

2 Related Work

Pivotal tokens. Abdin et al. [2024] introduce PTS to build preference pairs for Phi-4 post-training; Sharma [2025] released an open implementation and event datasets. Both use PTS as a data-generation step; the representational question—whether the property is present in activations beforehand—has not to our knowledge been asked, nor has the relationship between PTS labels and uncertainty been measured.

Uncertainty as the incumbent account. Wang et al. [2025] show that roughly 20% of tokens carry high entropy, act as branch points, and account for most of the gain from reinforcement learning with verifiable rewards. Section 5.2 is essentially a measurement of how much of PTS's criterion this account already covers.

Sentence-scale reasoning importance. Thought Anchors [Bogdan et al., 2025] identifies influential sentences by counterfactual resampling. We work at token scale, where the $t-1$ prediction problem is well posed: a sentence boundary is not a decoding step.

Probing and steering. Linear probes on intermediate representations [Alain and Bengio, 2018, Marks and Tegmark, 2023]; contrastive activation addition [Rimsky et al., 2024, Huang, 2024]; conditional steering [Lee et al., 2024]. Process reward models and value probes [Li et al., 2024] estimate $P(\text{success} \mid \text{prefix})$, a *level*; pivotality is $|\Delta p|$, a *sensitivity*.

3 Method

3.1 Generating the labels

We reimplement token-scale PTS with the same semantics as Sharma [2025] but express the bisection as a state machine, so that the rollouts of many questions can be batched together. The tree is sequential within a question—both children of a node need the node's own midpoint estimate—but questions are independent, and batching across them gives $8.9\times$ the generation throughput of running one node at a time. Emitted events carry the pivotal token's exact absolute position and the full searched rollout.

Two construction details materially affect the result.

Branches, not a canonical sequence. PTS samples several rollouts per question and they diverge after a few tokens. A construction that collapses a question to its longest rollout and matches the others against it as string prefixes silently discards every pivot on a diverging branch. On the released Qwen3-0.6B events, 104 questions carry 1,245 pivotal tokens over 447 distinct branches, and collapsing retains 318 of them—a 74% loss. We treat every maximal branch as its own sequence, and because branches share prefixes (and the residual stream at position p depends only on tokens $0 \dots p$) we label each distinct context exactly once.

Positions are exact, not inferred. Byte-pair encoding does not guarantee $\mathrm{enc}(a+b) = \mathrm{enc}(a)+\mathrm{enc}(b)$, so recovering a pivot's index by re-tokenizing its prefix is unsafe: a pivot token such as "." frequently merges with the preceding word. For data we generate this problem does not arise. For the released datasets, which predate the position field, we verify each recovered index against the reported token id and discard rows that fail.

3.2 Label noise in the oracle

$\widehat{\Delta p}$ is a difference of two binomial estimates from S rollouts each, accepted when $|\widehat{\Delta p}| \geq \tau$. Under the null of no true change $\mathrm{Var}(\widehat{\Delta p}) = 2p(1-p)/S$, so at the reference setting $S{=}50$, $\tau{=}0.2$ the acceptance threshold sits at only 2σ: a false-acceptance rate near 4.6% per tested position at $p{=}0.5$. We therefore store the raw counts $(n_{\text{before}}, s_{\text{before}}, n_{\text{after}}, s_{\text{after}})$ rather than a verdict, which makes τ a post-hoc knob and lets every event carry a Wilson interval. [TODO: report the accuracy gap between the full label set and a high-confidence subset.]

3.3 Probes, baselines, and how they are compared

Probes are ℓ_2-regularized logistic regressions on the residual stream, fit on standardized features and converted back to raw activation space so the resulting vector is usable as a direction. Layer and regularization strength are chosen on an inner split of the *training* questions; the test split is used once.

Baselines: token identity, as a smoothed log-odds lookup and as one-hot logistic regression, both fit on train and scored on held-out rows; next-token entropy, negative top-2 margin, and negative top-1 probability, computed from the model's own distribution at $t-1$; and a random unit direction.

Comparisons are paired. Confidence intervals are bootstrapped over *questions*, since positions within one reasoning trace are not independent. Crucially, when comparing the probe against a baseline we bootstrap the *difference* of AUROCs on the same resampled questions, rather than checking whether two marginal intervals overlap. The latter is badly conservative and would, on our data, have supported a conclusion the paired test rejects.

4 Experimental Setup

Model: Qwen3-0.6B [Qwen Team, 2025], 28 layers, $d{=}1024$. Task: GSM8K [Cobbe et al., 2021], conditioned on the bare question without a chat template, matching the released datasets and therefore without extended-thinking mode. Search: 3,000 candidate questions, $S{=}40$ rollouts per estimate, one searched rollout per question, $\tau{=}0.2$, 320 max new tokens, run on a single Blackwell-class GPU.

PTS only visits questions whose baseline success probability lies inside $[0.2, 0.8]$: a question the model always or never solves has no pivotal token, because a saturated probability cannot move. Of 3,000 candidates, 456 yielded at least one event, giving 688 pivotal positions and an equal number of sampled non-pivotal positions. Split by question: 365 train questions (1,132 rows) and 91 test questions (244 rows). Activations are read at all 29 residual-stream positions in bfloat16 at the labelled positions only.

5 Results

5.1 Pivotality is decodable, and is not token identity

At the selected layer ($L{=}22$) and penalty ($C{=}10^{-3}$), the probe reaches **0.855 AUROC** $[0.799, 0.903]$ at accuracy 0.775 on 244 held-out positions from 91 unseen questions. Table 1 reports every detector, and the paired comparison that matters.

Table 1: Predicting whether the *next* token is pivotal, from the residual stream at $t-1$. Differences are bootstrapped on the same resampled questions as the probe; $P(\Delta \leq 0)$ is the fraction of resamples in which the baseline matches or beats the probe.

Detector	AUROC	Probe − baseline	95% CI	$P(\Delta \leq 0)$
Linear probe	**0.855**	—	—	—
Token identity (freq. table)	0.662	$+0.191$	$[+0.116, +0.258]$	< 0.001
Token identity (one-hot LR)	0.696	—	—	—
Negative top-1 probability	0.823	$+0.030$	$[-0.047, +0.110]$	0.233
Negative top-2 margin	0.821	$+0.032$	$[-0.048, +0.115]$	0.227
Next-token entropy	0.814	$+0.039$	$[-0.037, +0.118]$	0.163
Mean-difference direction	0.820	—	—	—
Random direction	0.453	—	—	—

The probe exceeds token identity by 0.191 AUROC with an interval far from zero. So pivotality is present in the residual stream one step before the token that realizes it, and is not merely a property of which token is about to be emitted.

5.2 But it does not separate from next-token uncertainty

Every uncertainty statistic we tested lands between 0.814 and 0.823, and in each case the paired difference from the probe has a confidence interval containing zero. Entropy alone—one scalar, available without any probe—recovers most of what a trained linear readout on 1,024 dimensions recovers.

We do not think this is an artifact, and we think it is the more informative half of the paper. A token can only move $P(\text{success})$ if the continuation was not already determined; positions where the probability *can* move are close to positions where the model is uncertain. On that reading, PTS's criterion is substantially an uncertainty criterion wearing an outcome-shaped coat, and the Phi-4 pivotal-token line and the forking-token line [Wang et al., 2025] have been measuring overlapping phenomena with very different instruments — one costing tens of rollouts per position, the other one forward pass.

The practical implication is uncomfortable for the amortization story we set out to tell: if the goal is to find high-leverage positions cheaply, entropy is already cheap and already works. The remaining case for a probe is the part entropy cannot do in principle, which brings us to the sign.

5.3 The sign is not decodable at this scale

Pivotality as defined in Eq. 1 is symmetric: a token that rescues an answer and one that ruins it get the same label. The sharper question is whether the *sign* is decodable at $t-1$—whether the model represents, one step early, that it is about to make a mistake. Entropy is sign-blind by construction and cannot answer it, so a positive result here would be immune to Section 5.2.

Restricting to pivotal positions and relabelling by $\text{sign}(\Delta p)$ gives 566 training and 122 test rows (56.5% helpful). A probe reaches 0.614 AUROC, with a 95% interval of $[0.482, 0.735]$ that contains chance. The signed mean-difference direction reaches 0.567 and token identity 0.555. We report this as a negative result. An earlier version of this analysis, on a smaller and noisier label set, gave 0.717 and looked promising; it did not survive a cleaner dataset with four times as many held-out questions.

5.4 No layer is special, and selection inflates the headline

Simulating a null in which all 29 layers are equally informative, each measured on 244 rows, gives an expected maximum close to the observed one, so the identity of the "best" layer carries little information; the signal is distributed across the mid-to-late stack.

The cost of getting this wrong is measurable. Choosing the layer on the reporting split gives 0.866; choosing it on training data and reporting on test gives 0.855. Under our earlier, leakier protocol the same gap was 0.063 AUROC—large enough to manufacture a result.

5.5 Regularization

AUROC moves from 0.800 at $C{=}1$ to 0.855 at $C{=}10^{-3}$, and the probe direction rotates toward the mean-difference direction as the penalty tightens (cosine 0.203 to 0.800). A weakly regularized probe still classifies but does so along a nearly orthogonal direction, which matters for anyone intending to use the probe's weights as an intervention direction rather than only as a classifier.

6 Limitations

A single 0.6B model on a single task family, with 91 held-out questions. The oracle's labels are themselves noisy (Section 3.2) in a way we have quantified analytically but not yet empirically. The acceptance filter retains only mid-difficulty questions, so pivot density is confounded with task difficulty relative to model capability and this label set is not a uniform sample of GSM8K. Conditioning on the bare question puts an instruct model off distribution; we observed rollouts in which the model hallucinated multiple-choice options, and at least one event whose extreme Δp traces to the answer-extraction cascade matching such a hallucination rather than to a genuine shift. Linear probes test linear decodability only. Everything here is correlational: we have not shown that the direction the probe identifies is causally used.

Most importantly, Section 5.2 is a null result on a comparison, not a proof of equivalence: 91 questions cannot rule out a real 4-point advantage. [TODO: Scale the question count and re-test; also re-run this comparison on the released label set with the fixed pipeline, to separate "PTS is an uncertainty criterion" from "our generation regime made it one".]

7 Conclusion

Pivotality is linearly present in a model's residual stream one position before the token that realizes it, at 0.855 AUROC, and is clearly separable from token identity. It is not separable from next-token uncertainty, which alone reaches 0.814—suggesting that an expensive, outcome-grounded, rollout-based criterion is largely recovering something a single forward pass already exposes. Whether the *direction* of the impending shift carries information beyond that remains open; at this sample size it does not.

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A Reproduction

Events: `stvngo/Qwen3-0.6B-pts-tokens`. Activations: `stvngo/Qwen3-0.6B-pivotal-activations`. All numbers in Section 5 come from `artifacts/probes_v2_full/qwen3-0.6b_report.json`, produced by `scripts/pts_run.py`, `scripts/build_and_extract.py` and `scripts/evaluate_probes_v2.py`. [TODO: Add the layer sweep and regularization-path figures.]